Code.org: Core Coding Concepts

The big ideas behind every program, explained simply.

Code.org | Beginner | Ages 6–14

💡 The Big Idea

- Every program, from a puzzle to a big game, is built from a few core ideas.
- Learn these ideas once and they work in every coding language.
- Code.org puzzles teach each one step by step.
- You can practise all of them without a computer too.

🧱 Sequence and Algorithms

- A sequence is a list of steps in the right order.
- An algorithm is a clear set of steps to solve a problem.
- Changing the order of the steps changes the result.
- Making a sandwich or brushing your teeth is an algorithm.

🔁 Loops and Events

- A loop repeats steps so you do not write them again and again.
- An event is something that happens, like a click, a key press or a touch.
- Code can wait for an event and then run a block.
- Loops and events are used in almost every game.

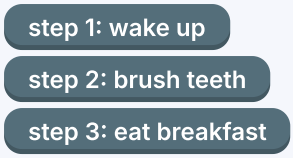
🔀 Conditionals, Variables and Functions

- A conditional says: if this is true, do that, otherwise do something else.
- A variable stores a value that can change, like a score.
- A function packs steps into one named block you can use many times.
- Breaking a big problem into small parts is called decomposition.

💻 Examples

Sequence

Code.org blocks

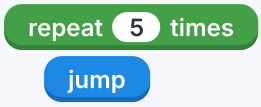


▶ What happens:

The steps happen in this order. Swapping them changes the result.

Loop

Code.org blocks

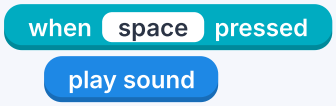


▶ What happens:

The character jumps 5 times without needing 5 separate jump blocks.

Event

Code.org blocks

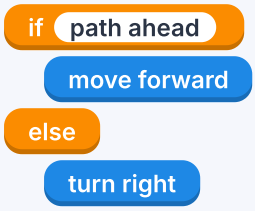


▶ What happens:

Nothing happens until the player presses the space key.

Conditional

Code.org blocks

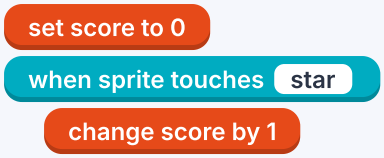


▶ What happens:

The character walks while there is a path. When there is none, it turns right.

Variable

Code.org blocks

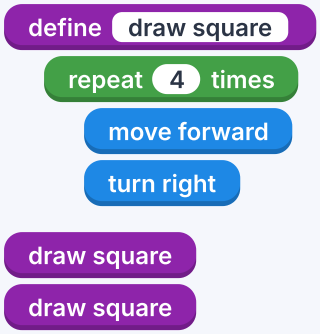


▶ What happens:

The score starts at 0 and goes up by 1 for every star collected.

Function

Code.org blocks



▶ What happens:

The function draws a square. Calling it twice draws two squares.

📚 Key Terms

Sequence
Steps in order.

Variable
A named box for a value.

Algorithm
A set of steps that solves a problem.

Function
A reusable block of steps.

Loop
Repeats steps.

Decomposition
Splitting a big problem into small ones.

Event
Something that triggers code.

Debugging
Finding and fixing mistakes.

Conditional
A choice: if this, then that.

Unplugged
Learning to code without a computer.

🧮 Quick Reference

```
Sequence     -> do things in order
```

```
Loop         -> repeat (n) times
Event        -> when [something] happens
Conditional  -> if / else
Variable     -> set score to 0
Function     -> define, then call it
```

✏️ Worked Examples

Q: Which concept lets you draw a square with 3 blocks instead of 8?
A: A loop. repeat 4 times: move forward, turn right.

Q: A game should end when lives reach 0. Which concept is that?
A: A conditional. If lives = 0 then end the game.

Q: You use the same 6 blocks in 5 places. How can you tidy it?
A: Make a function with those 6 blocks and call it 5 times.

🚫 Common Mistakes

- Mixing up the order of the steps in a sequence.
- Using the wrong number in a loop, like repeating 3 times instead of 4.
- Forgetting the else path in a conditional.
- Copying blocks again and again instead of using a loop or a function.
- Giving up on a puzzle instead of testing one change at a time.

👨‍👩‍👧 Parent Corner

- Play Robot: your child gives you step by step directions to make a drink. Do exactly what they say.
- Ask them to find loops in daily life, like brushing every tooth.
- Ask What if? questions to practise conditionals: What if it rains?
- Let them explain a function: What did you pack into one block?
- Celebrate trying different ideas.

🎯 Quick Check

1. What is a sequence?

✅ A list of steps in the right order.

2. What does a loop do?

✅ It repeats steps.

3. What is an event?

✅ Something that happens, like a click, that triggers code.

4. What does a conditional do?

✅ It lets the program choose, using if and else.

5. Why do we use functions?

✅ To reuse a group of blocks without writing them again.